The Geometry of Linear Regression

Lecture Notes: Seeing OLS Through the Projection Theorem

Statistical Machine Learning Lab

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Abstract

These lecture notes develop ordinary least squares (OLS) regression entirely from the perspective of the *Projection Theorem* in inner product spaces. Every classical result—the normal equations, variance decomposition, R^2, the Gauss–Markov theorem, the Frisch–Waugh–Lovell theorem, leverage and influence diagnostics, multicollinearity, omitted variable bias, Ridge and LASSO regularisation, the F-test, and confidence ellipsoids—is derived as a geometric consequence of orthogonal projection onto the column space of the design matrix. The unifying principle: *regression is a shadow cast by the data vector onto the column space, and the residual is the perpendicular distance from object to shadow.*

Contents

1 Prologue: One Theorem, Ten Perspectives

The Flashlight Metaphor

Imagine a flashlight shining straight down onto a flat wall. The **object** is the data vector $\boldsymbol{y} \in \mathbb{R}^n$. The **wall** is the column space $\mathcal{C}(\boldsymbol{X})$. The **shadow** is the fitted vector $\hat{\boldsymbol{y}}$. The **gap** between object and shadow is the residual $\boldsymbol{e}$. The shadow always falls at a **right angle**—the residual is perpendicular to the wall.

Every result in these notes flows from a single theorem.

Theorem 1.1 (The Projection Theorem)**.** *Let S be a closed subspace of an inner product space V, and let $\boldsymbol{y} \in V$. There exists a unique $\hat{\boldsymbol{y}} \in S$ such that*

$$\|\boldsymbol{y} - \hat{\boldsymbol{y}}\| \;\leq\; \|\boldsymbol{y} - \boldsymbol{s}\| \qquad \textit{for all } \boldsymbol{s} \in S.$$

Moreover, $\hat{\boldsymbol{y}}$ is characterised by the orthogonality condition

$$(\boldsymbol{y} - \hat{\boldsymbol{y}}) \perp S,$$

i.e. $\langle \boldsymbol{y} - \hat{\boldsymbol{y}}, \boldsymbol{s} \rangle = 0$ for every $\boldsymbol{s} \in S$.

Ordinary least squares is the Projection Theorem applied to $V = \mathbb{R}^n$ with the Euclidean inner product and $S = \mathcal{C}(\boldsymbol{X})$.

2 Where Coefficients Come From

2.1 Setup and notation

We observe n data points (x_i, y_i), $i = 1, \ldots, n$. Stack the responses into a single vector $\boldsymbol{y} = (y_1, \ldots, y_n)^\top \in \mathbb{R}^n$ and form the $n \times p$ **design matrix**

$$\boldsymbol{X} = \begin{pmatrix} \boldsymbol{x}_1^\top \\ \vdots \\ \boldsymbol{x}_n^\top \end{pmatrix} \in \mathbb{R}^{n \times p},$$

where typically the first column is a column of ones (the intercept). The **column space** of $\boldsymbol{X}$ is

$$\mathcal{C}(\boldsymbol{X}) = \{\boldsymbol{X}\boldsymbol{\beta} : \boldsymbol{\beta} \in \mathbb{R}^p\} \;\subseteq\; \mathbb{R}^n.$$

Every vector in $\mathcal{C}(\boldsymbol{X})$ represents the set of fitted values for *some* choice of coefficients.

2.2 OLS as orthogonal projection

Definition 2.1 (OLS estimator)**.** The OLS estimator is

$$\hat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta} \in \mathbb{R}^p}{\arg\min} \, \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2 .$$

Since $\mathcal{C}(\boldsymbol{X})$ is a closed subspace of $\mathbb{R}^n$, the Projection Theorem (**??**) guarantees a unique minimiser. The fitted values $\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}}$ are the orthogonal projection of $\boldsymbol{y}$ onto $\mathcal{C}(\boldsymbol{X})$, and the residual $\boldsymbol{e} = \boldsymbol{y} - \hat{\boldsymbol{y}}$ is perpendicular to $\mathcal{C}(\boldsymbol{X})$.

Key Geometric Insight

$\hat{\boldsymbol{\beta}}$ is the coordinate vector of the foot of the perpendicular from $\boldsymbol{y}$ to $\mathcal{C}(\boldsymbol{X})$. The coefficients are not arbitrary numbers from a formula—they locate the point on the column space closest to the data.

2.3 The normal equations

The orthogonality condition $\boldsymbol{e} \perp \mathcal{C}(\boldsymbol{X})$ means $\langle \boldsymbol{X}\boldsymbol{v}, \boldsymbol{e}\rangle = 0$ for all $\boldsymbol{v} \in \mathbb{R}^p$, i.e.

$$\boldsymbol{X}^\top \boldsymbol{e} = \boldsymbol{0} \qquad \Longleftrightarrow \qquad \boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}) = \boldsymbol{0} \qquad \Longleftrightarrow \qquad \boxed{\boldsymbol{X}^\top \boldsymbol{X}\,\hat{\boldsymbol{\beta}} = \boldsymbol{X}^\top \boldsymbol{y}.}$$

These are the **normal equations**. If $\boldsymbol{X}$ has full column rank, $\boldsymbol{X}^\top\boldsymbol{X}$ is invertible and

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{y}. \tag{1}$$

The formula is a recipe for dropping a perpendicular.

2.4 Building intuition: from $\mathbb{R}^2$ to $\mathbb{R}^n$

- With $n = 2$ observations, $\boldsymbol{y} \in \mathbb{R}^2$ and the column space is a line through the origin (for simple regression with intercept, a line in the plane).
- With $n = 3$, $\boldsymbol{y} \in \mathbb{R}^3$ and a model with intercept and one predictor spans a 2-dimensional plane—the geometry is fully visible.
- For $n = 2000$, the geometry is identical; only the ambient dimension changes. The right angle between $\boldsymbol{e}$ and $\mathcal{C}(\boldsymbol{X})$ holds in every dimension.

3 The Hat Matrix and the Right Angle

3.1 The hat matrix

Definition 3.1. The **hat matrix** (or projection matrix) is

$$\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top.$$

It "puts the hat on $\boldsymbol{y}$": $\hat{\boldsymbol{y}} = \boldsymbol{H}\boldsymbol{y}$.

Proposition 3.2 (Properties of $\boldsymbol{H}$)**.**
(i) ***Idempotent:*** $\boldsymbol{H}^2 = \boldsymbol{H}$.
(ii) ***Symmetric:*** $\boldsymbol{H}^\top = \boldsymbol{H}$.
(iii) $\operatorname{rank}(\boldsymbol{H}) = \operatorname{tr}(\boldsymbol{H}) = p$.
(iv) $0 \le h_{ii} \le 1$ *for every diagonal entry, and* $\sum_i h_{ii} = p$.

Proof. (i) $\boldsymbol{H}^2 = \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top = \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top = \boldsymbol{H}$. (ii) Follows directly from the symmetry of $(\boldsymbol{X}^\top\boldsymbol{X})^{-1}$. (iii)–(iv) follow from idempotency: the eigenvalues of an idempotent matrix are 0 or 1, so $\operatorname{tr}(\boldsymbol{H}) = \operatorname{rank}(\boldsymbol{H}) = p$, and each diagonal entry $h_{ii} = \boldsymbol{e}_i^\top\boldsymbol{H}\boldsymbol{e}_i$ lies in $[0, 1]$ because $\boldsymbol{H}$ is positive semi-definite with eigenvalues in $\{0, 1\}$. □

Geometric meaning. Idempotency says: *the shadow of a shadow is itself.* If $\hat{\boldsymbol{y}}$ is already on the column space, projecting it again changes nothing. Symmetry says the projection treats all directions equally—it flattens onto the plane without distortion.

3.2 The annihilator matrix

Define $\boldsymbol{M} = \boldsymbol{I}_n - \boldsymbol{H}$. Then $\boldsymbol{e} = \boldsymbol{M}\boldsymbol{y}$, and $\boldsymbol{M}$ inherits the same properties: $\boldsymbol{M}^2 = \boldsymbol{M}$, $\boldsymbol{M}^\top = \boldsymbol{M}$. It is the projection onto the *orthogonal complement* $\mathcal{C}(\boldsymbol{X})^\perp$. Together, $\boldsymbol{H}$ and $\boldsymbol{M}$ decompose any vector into two perpendicular components:

$$\boldsymbol{y} = \underbrace{\boldsymbol{H}\boldsymbol{y}}_{\hat{\boldsymbol{y}} \in \mathcal{C}(\boldsymbol{X})} + \underbrace{\boldsymbol{M}\boldsymbol{y}}_{\boldsymbol{e} \in \mathcal{C}(\boldsymbol{X})^\perp}.$$

Warning

The orthogonality $\boldsymbol{X}^\top \boldsymbol{e} = \boldsymbol{0}$ is a property of the *method*, not a validation of the *model*. You could regress salary on zodiac sign and the residuals would still be perpendicular to the zodiac column. Residual-predictor uncorrelatedness is by construction.

4 Pythagoras Runs a Regression

4.1 Variance decomposition as the Pythagorean theorem

Since $\hat{\boldsymbol{y}} \perp \boldsymbol{e}$ and $\boldsymbol{y} = \hat{\boldsymbol{y}} + \boldsymbol{e}$, the Pythagorean theorem gives

$$\|\boldsymbol{y}\|^2 = \|\hat{\boldsymbol{y}}\|^2 + \|\boldsymbol{e}\|^2 . \tag{2}$$

Working with demeaned vectors $\tilde{\boldsymbol{y}} = \boldsymbol{y} - \bar{y}\boldsymbol{1}$ (which is the projection of $\boldsymbol{y}$ onto the orthogonal complement of the intercept), we obtain the fundamental variance decomposition. Before stating it, let us define the three sums of squares.

Definition 4.1 (Sums of Squares)**.**

- **SST** (Total Sum of Squares) $= \sum_{i=1}^n (y_i - \bar{y})^2 = \|\tilde{\boldsymbol{y}}\|^2$.
  This measures the *total variability* in the response. It answers: "How spread out are the y_i values around their mean?"
- **SSR** (Regression Sum of Squares, also called the "explained" sum of squares) $= \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \left\|\tilde{\hat{\boldsymbol{y}}}\right\|^2$.
  This measures the variability *explained by the model*. It answers: "How much of the spread in $\boldsymbol{y}$ is captured by the fitted values?"
- **SSE** (Error Sum of Squares, also called the "residual" sum of squares) $= \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \|\boldsymbol{e}\|^2$.
  This measures the variability *left unexplained* by the model. It answers: "How far are the actual values from the predictions?"

With these definitions, the Pythagorean theorem gives us:

$$\boxed{\underbrace{\|\tilde{\boldsymbol{y}}\|^2}_{\text{SST}} = \underbrace{\left\|\tilde{\hat{\boldsymbol{y}}}\right\|^2}_{\text{SSR}} + \underbrace{\|\boldsymbol{e}\|^2}_{\text{SSE}}.}$$

In words: the total variability in the data equals the variability explained by the regression plus the variability left over in the residuals. The variance decomposition *is* the Pythagorean theorem applied to the right triangle $(\tilde{\boldsymbol{y}},\ \tilde{\hat{\boldsymbol{y}}},\ \boldsymbol{e})$ in $\mathbb{R}^n$.

4.2 R^2 is the cosine of an angle

The coefficient of determination R^2 is usually introduced via the formula $R^2 = 1 - \text{SSE}/\text{SST}$, or equivalently $R^2 = \text{SSR}/\text{SST}$. Let us derive this from the geometry of the regression right triangle.

Look at the right triangle in the figure above. The *hypotenuse* is the demeaned data vector $\tilde{\boldsymbol{y}}$ (with length $\sqrt{\text{SST}}$), the *adjacent side* is the demeaned fitted vector $\tilde{\hat{\boldsymbol{y}}}$ (with length $\sqrt{\text{SSR}}$), and the *opposite side* is the residual $\boldsymbol{e}$ (with length $\sqrt{\text{SSE}}$). The angle θ sits at the origin, between the hypotenuse and the adjacent side.

From basic trigonometry, the cosine of an angle in a right triangle is

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{\left\|\tilde{\hat{\boldsymbol{y}}}\right\|}{\|\tilde{\boldsymbol{y}}\|} = \frac{\sqrt{\text{SSR}}}{\sqrt{\text{SST}}}.$$

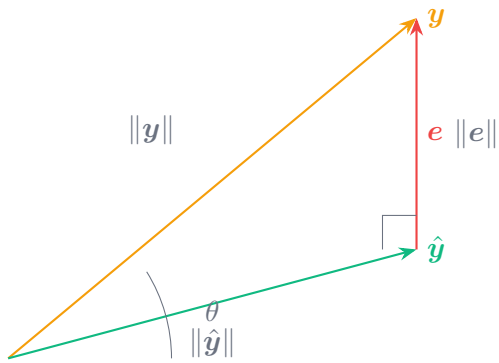

Figure 1: The regression right triangle. The Pythagorean theorem gives SST = SSR + SSE.

Squaring both sides:

$$\cos^2\theta = \frac{\left\|\tilde{\hat{\boldsymbol{y}}}\right\|^2}{\|\tilde{\boldsymbol{y}}\|^2} = \frac{\text{SSR}}{\text{SST}}.$$

But SSR/SST is precisely the definition of R^2, so we arrive at

$$\boxed{R^2 = \cos^2\theta = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}.} \tag{3}$$

The last equality follows directly from SST = SSR + SSE.

Key Geometric Insight

R^2 is literally the squared cosine of the angle between the data vector and its projection.

- When $\theta \approx 0$, the data vector nearly lies on the column space. The shadow almost coincides with the object, meaning the model explains nearly all the variability: $R^2 \approx 1$.
- When $\theta \approx 90$ř, the data vector is nearly perpendicular to the column space. The shadow is tiny—the model explains almost nothing: $R^2 \approx 0$.
- $R^2 = 0.75$ means $\cos^2\theta = 0.75$, so $\theta \approx 30$ř. Three-quarters of the data's variability is captured by the projection; one quarter is residual.

4.3 The Gauss–Markov theorem: OLS is the closest point

Theorem 4.2 (Gauss–Markov)**.** *Among all linear unbiased estimators of $\boldsymbol{\beta}$, OLS has the smallest variance (in the Loewner ordering). That is, $\hat{\boldsymbol{\beta}}$ is BLUE (Best Linear Unbiased Estimator).*

Geometric proof sketch. Any linear unbiased estimator $\tilde{\boldsymbol{\beta}} = \boldsymbol{C}\boldsymbol{y}$ with $\boldsymbol{C}\boldsymbol{X} = \boldsymbol{I}_p$ produces fitted values $\boldsymbol{X}\tilde{\boldsymbol{\beta}} \in \mathcal{C}(\boldsymbol{X})$. But $\hat{\boldsymbol{y}} = \boldsymbol{H}\boldsymbol{y}$ is the *closest point* in $\mathcal{C}(\boldsymbol{X})$ to $\boldsymbol{y}$ by the Projection Theorem. Any other point in the column space is further away, hence captures more noise variance.

Formally, write $\boldsymbol{C} = (\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top + \boldsymbol{D}$ where $\boldsymbol{D}\boldsymbol{X} = \boldsymbol{0}$. Then $\text{Var}(\tilde{\boldsymbol{\beta}}) = \sigma^2(\boldsymbol{C}\boldsymbol{C}^\top) = \sigma^2[(\boldsymbol{X}^\top\boldsymbol{X})^{-1} + \boldsymbol{D}\boldsymbol{D}^\top] \succeq \sigma^2(\boldsymbol{X}^\top\boldsymbol{X})^{-1} = \text{Var}(\hat{\boldsymbol{\beta}})$. □

4.4 R^2 monotonicity

Proposition 4.3. *Adding a predictor to $\boldsymbol{X}$ can only increase (or maintain) R^2.*

Proof. Adding a column to $\boldsymbol{X}$ enlarges $\mathcal{C}(\boldsymbol{X})$. A projection onto a *larger* subspace is at least as close to $\boldsymbol{y}$ as a projection onto a smaller one, so $\|\boldsymbol{e}\|^2$ cannot increase, hence $R^2 = 1 - \|\boldsymbol{e}\|^2/\text{SST}$ cannot decrease. □

Warning

R^2 monotonicity is a geometric fact about the *training sample*. It says nothing about out-of-sample prediction. Adding polynomial terms $(x, x^2, \ldots, x^{12})$ drives R^2 toward 1 while destroying predictive performance—the shadow gets closer in this room but means nothing for the next room.

5 Peeling Apart the Variables: Frisch–Waugh–Lovell

5.1 What "controlling for" means geometrically

What does "conditioning on" or "controlling for" a variable mean?

In everyday regression language, you will hear phrases like "the effect of education on income, *controlling for* age" or "the coefficient of x_2 *conditional on* x_1." For an undergraduate, this can sound mysterious. Here is the plain-English idea:

Suppose you want to know whether education affects income. But older people tend to have both more education and higher income simply because they have been working longer. If you just regress income on education, you cannot tell whether the apparent effect comes from education itself or from age lurking in the background. *Controlling for age* means: compare people *of the same age* and then ask whether more education is still associated with higher income. In other words, you strip out the age-related variation from both income and education, and then look at what remains.

In a regression with multiple predictors, this is *exactly* what the coefficient of a predictor measures. When we write $\boldsymbol{y} = \boldsymbol{X}_1\boldsymbol{\beta}_1 + \boldsymbol{X}_2\boldsymbol{\beta}_2 + \boldsymbol{e}$, the coefficient $\hat{\boldsymbol{\beta}}_2$ answers: *what is the effect of* $\boldsymbol{X}_2$ *after removing ("partialling out") the influence of* $\boldsymbol{X}_1$*?* The Frisch–Waugh–Lovell theorem below makes this operation precise.

Theorem 5.1 (Frisch–Waugh–Lovell)**.** *Partition the design matrix as* $\boldsymbol{X} = [\boldsymbol{X}_1 \;\; \boldsymbol{X}_2]$ *with conformable coefficient vector* $\boldsymbol{\beta} = (\boldsymbol{\beta}_1^\top, \boldsymbol{\beta}_2^\top)^\top$*. Let* $\boldsymbol{M}_1 = \boldsymbol{I} - \boldsymbol{X}_1(\boldsymbol{X}_1^\top\boldsymbol{X}_1)^{-1}\boldsymbol{X}_1^\top$ *be the annihilator for* $\mathcal{C}(\boldsymbol{X}_1)$*. Then*

$$\hat{\boldsymbol{\beta}}_2 = (\boldsymbol{X}_2^\top\boldsymbol{M}_1\boldsymbol{X}_2)^{-1}\boldsymbol{X}_2^\top\boldsymbol{M}_1\boldsymbol{y}. \tag{4}$$

That is, $\hat{\boldsymbol{\beta}}_2$ *from the full regression equals the OLS coefficient from regressing* $\boldsymbol{M}_1\boldsymbol{y}$ *on* $\boldsymbol{M}_1\boldsymbol{X}_2$*—the* residualised *response on the* residualised *predictors.*

Proof. The normal equations in block form are

$$\begin{pmatrix} \boldsymbol{X}_1^\top\boldsymbol{X}_1 & \boldsymbol{X}_1^\top\boldsymbol{X}_2 \\ \boldsymbol{X}_2^\top\boldsymbol{X}_1 & \boldsymbol{X}_2^\top\boldsymbol{X}_2 \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\beta}}_1 \\ \hat{\boldsymbol{\beta}}_2 \end{pmatrix} = \begin{pmatrix} \boldsymbol{X}_1^\top\boldsymbol{y} \\ \boldsymbol{X}_2^\top\boldsymbol{y} \end{pmatrix}.$$

Solve the first block for $\hat{\boldsymbol{\beta}}_1$ and substitute into the second. After simplification (using $\boldsymbol{M}_1 = \boldsymbol{I} - \boldsymbol{X}_1(\boldsymbol{X}_1^\top\boldsymbol{X}_1)^{-1}\boldsymbol{X}_1^\top$ and the identity $\boldsymbol{M}_1\boldsymbol{X}_1 = \boldsymbol{0}$), one obtains (**??**). □

Key Geometric Insight

"Controlling for $\boldsymbol{X}_1$" is a geometric operation: project $\boldsymbol{y}$ and $\boldsymbol{X}_2$ onto the orthogonal complement of $\mathcal{C}(\boldsymbol{X}_1)$, then perform simple regression in that cleaned-up space. The multiple regression coefficient is the simple regression coefficient in the residualised subspace.

The geometric analogy: two-step projection. Picture a three-dimensional room. The data vector $\boldsymbol{y}$ is a point floating in the room. The full column space $\mathcal{C}(\boldsymbol{X}_1, \boldsymbol{X}_2)$ is a plane (the "floor"). The subspace $\mathcal{C}(\boldsymbol{X}_1)$ is a line within that floor.

The FWL theorem says you can find the regression coefficient $\hat{\boldsymbol{\beta}}_2$ via two sequential projections instead of one joint projection:

1. **Step 1: Remove $\boldsymbol{X}_1$'s influence.** Project both $\boldsymbol{y}$ and $\boldsymbol{X}_2$ onto the space *perpendicular* to $\mathcal{C}(\boldsymbol{X}_1)$. This is like "looking at the room from directly along the $\boldsymbol{X}_1$ line"—the line collapses to a point, and only the variation perpendicular to $\boldsymbol{X}_1$ remains visible. The residualised vectors $\boldsymbol{M}_1\boldsymbol{y}$ and $\boldsymbol{M}_1\boldsymbol{X}_2$ live in this reduced space.
2. **Step 2: Regress in the cleaned space.** Now perform a simple regression of $\boldsymbol{M}_1\boldsymbol{y}$ on $\boldsymbol{M}_1\boldsymbol{X}_2$ in the reduced space. The resulting coefficient is exactly $\hat{\boldsymbol{\beta}}_2$ from the full multiple regression.

This two-step procedure is not an approximation—it gives the *exact same* coefficient as the full regression. Geometrically, the full projection onto the plane can be decomposed into a projection onto the line, followed by a projection of the remainder onto the perpendicular direction within the plane.

5.2 The effect of predictor correlation

The FWL theorem reveals *why* correlated predictors cause trouble. To understand this, think carefully about what the residualisation step does.

The residualised predictor and its length. When we "control for" $\boldsymbol{X}_1$, we replace $\boldsymbol{X}_2$ by its residuals after regressing on $\boldsymbol{X}_1$:

$$\boldsymbol{M}_1\boldsymbol{X}_2 = \boldsymbol{X}_2 - \underbrace{\boldsymbol{X}_1(\boldsymbol{X}_1^\top\boldsymbol{X}_1)^{-1}\boldsymbol{X}_1^\top\boldsymbol{X}_2}_{\text{part of } \boldsymbol{X}_2 \text{ explained by } \boldsymbol{X}_1}.$$

The residualised predictor $\boldsymbol{M}_1\boldsymbol{X}_2$ contains only the "unique information" in $\boldsymbol{X}_2$—the part that $\boldsymbol{X}_1$ cannot account for.

Now think about what happens as the correlation between $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$ changes:

- **Uncorrelated predictors** ($\boldsymbol{X}_1^\top\boldsymbol{X}_2 \approx \boldsymbol{0}$): The projection of $\boldsymbol{X}_2$ onto $\mathcal{C}(\boldsymbol{X}_1)$ is nearly zero, so $\boldsymbol{M}_1\boldsymbol{X}_2 \approx \boldsymbol{X}_2$. Almost all of $\boldsymbol{X}_2$'s information is unique. The residualised predictor is long, and the coefficient is estimated precisely.
- **Moderately correlated predictors**: Part of $\boldsymbol{X}_2$ is "used up" by $\boldsymbol{X}_1$. The residualised predictor $\boldsymbol{M}_1\boldsymbol{X}_2$ is shorter. Less unique information remains, and the coefficient estimate becomes noisier.
- **Highly correlated predictors** ($\boldsymbol{X}_2 \approx \boldsymbol{X}_1\boldsymbol{c}$ for some $\boldsymbol{c}$): Nearly all of $\boldsymbol{X}_2$ is explained by $\boldsymbol{X}_1$. The residualised predictor $\boldsymbol{M}_1\boldsymbol{X}_2$ is very short—close to the zero vector. There is almost no unique information left. The coefficient estimate becomes wildly unstable.

Geometric picture. Imagine $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$ as two vectors in $\mathbb{R}^n$. Residualising $\boldsymbol{X}_2$ with respect to $\boldsymbol{X}_1$ means projecting $\boldsymbol{X}_2$ onto the plane perpendicular to $\boldsymbol{X}_1$ and keeping only the perpendicular component. If $\boldsymbol{X}_2$ points in nearly the same direction as $\boldsymbol{X}_1$, this perpendicular component is tiny. You are trying to estimate a coefficient by regressing on a near-zero vector, which amplifies any noise enormously.

The standard error formula. The standard error of $\hat{\boldsymbol{\beta}}_2$ is

$$\mathrm{SE}(\hat{\beta}_{2,j}) = \sqrt{\frac{s^2}{\|\boldsymbol{M}_1\boldsymbol{x}_{2,j}\|^2}},$$

where $s^2 = \|\boldsymbol{e}\|^2/(n-p)$ is the estimated residual variance. As correlation increases, $\|\boldsymbol{M}_1\boldsymbol{x}_{2,j}\| \to 0$ and the denominator vanishes, sending $\mathrm{SE} \to \infty$. This is the geometric root of *multicollinearity*: the coefficient is not wrong on average (it is still unbiased), but it is estimated so imprecisely that it becomes practically useless.

6 Leverage and Influence

6.1 Leverage as geometric distance

Each fitted value $\hat{y}_i = \sum_j h_{ij} y_j$ is a weighted combination of all responses. The diagonal element h_{ii} measures how much observation i influences *its own* prediction.

Definition 6.1. The **leverage** of observation i is h_{ii}, the i-th diagonal element of $\boldsymbol{H}$. By Proposition **??**(iv), $1/n \leq h_{ii} \leq 1$ and $\sum_i h_{ii} = p$. The conventional threshold for "high leverage" is $h_{ii} > 2p/n$.

Geometrically, h_{ii} measures how far the i-th predictor vector $\boldsymbol{x}_i^\top$ lies from the centroid of the predictor space. Observations in unusual predictor combinations have long "arms" in the tug-of-war that determines the projection.

6.2 Cook's distance: leverage × surprise

High leverage means an observation *can* distort the projection. Whether it *does* depends on whether it also has a large residual. Cook's distance combines both pieces of information into a single measure of *influence*.

The thought experiment behind Cook's distance. Imagine deleting observation i from the dataset and re-running the entire regression. How much do the fitted values change? If they barely move, observation i is not influential. If they change dramatically, observation i is pulling the regression toward itself.

Cook's distance measures exactly this: it is proportional to the squared distance between the fitted values from the full regression and the fitted values with observation i removed:

$$D_i \propto \left\| \hat{\boldsymbol{y}} - \hat{\boldsymbol{y}}_{(i)} \right\|^2,$$

where $\hat{\boldsymbol{y}}_{(i)}$ denotes the fitted values from the regression that excludes observation i. The remarkable fact is that you do not actually need to refit the regression n times. Using the hat matrix, Cook's distance can be computed in closed form.

Definition 6.2. Cook's distance for observation i is

$$D_i = \frac{e_i^2}{p \cdot \mathrm{MSE}} \cdot \frac{h_{ii}}{(1 - h_{ii})^2},$$

where $\mathrm{MSE} = \|\boldsymbol{e}\|^2 / (n - p)$ is the mean squared error.

Reading the formula. Cook's distance is a product of two factors:

1. $\frac{e_i^2}{p \cdot \mathrm{MSE}}$ — this is the **surprise factor**. It measures how large the residual of observation i is, relative to the overall noise level. A large residual means the observation does not fit the current model well; it "wants" the regression plane to tilt toward it.
2. $\frac{h_{ii}}{(1-h_{ii})^2}$ — this is the **leverage factor**. It measures how much "mechanical advantage" the observation has. An observation at an extreme predictor value (h_{ii} close to 1) can exert enormous force on the regression; an observation near the center of the predictor cloud (h_{ii} close to $1/n$) has little leverage, no matter how unusual its y-value.

An analogy. Think of the regression line as a seesaw (lever) balanced on the centroid of the data. *Leverage* (h_{ii}) is how far from the fulcrum observation i sits—its position along the seesaw. The *residual* (e_i) is the force it pushes with—how far above or below the seesaw it is. Cook's distance is the resulting *torque*: force times distance from the fulcrum. A person sitting at the end of the seesaw (high leverage) and pushing hard (large residual) tilts the seesaw a lot. A person sitting near the fulcrum (low leverage), no matter how hard they push, barely tilts it.

When to worry. A common rule of thumb is to investigate observations with $D_i > 4/n$ or $D_i > 1$. But the real message is qualitative: look at observations with large Cook's distance and ask whether they represent data errors, unusual subpopulations, or genuine outliers that should influence the model.

Key Geometric Insight

Influence = leverage × surprise. An observation needs *both* a long arm (h_{ii} large) *and* a strong pull ($|e_i|$ large) to actually move the regression. A high-leverage observation that fits the trend is harmless (large h_{ii}, small e_i, so D_i is small). A central observation with a large residual has limited impact (small h_{ii}, large e_i, so D_i is moderate). The dangerous case is an observation that is *both* extreme in its predictor values *and* does not follow the pattern of the other data.

7 When the Geometry Fails

7.1 Multicollinearity: the geometry breaks

When predictors are nearly linearly dependent, $\mathcal{C}(\boldsymbol{X})$ degenerates from a "thick" plane into a paper-thin sliver. Let us understand why this is dangerous and what exactly goes wrong.

What multicollinearity means. Multicollinearity occurs when one predictor can be approximately written as a linear combination of others. For example, if $x_2 \approx 3x_1 + 2x_3$, then the three predictors contain nearly the same information. The design matrix $\boldsymbol{X}$ is close to being rank-deficient.

Why it is dangerous: the pancake analogy. Consider fitting a model with two highly correlated predictors. Geometrically, the column space $\mathcal{C}(\boldsymbol{X})$ is a 2-dimensional plane in $\mathbb{R}^n$. When the two predictors are nearly proportional, this plane is not a "square" plane—it is extremely elongated, like a thin pancake or a nearly 1-dimensional line.

Now imagine projecting the data vector $\boldsymbol{y}$ onto this thin pancake. The projection itself is fine: $\hat{\boldsymbol{y}}$ lands on the pancake and $\boldsymbol{e}$ is perpendicular. The *total* prediction $\hat{\boldsymbol{y}}$ is stable. But the *individual coefficients* are not: there are many different ways to decompose $\hat{\boldsymbol{y}}$ into contributions along two nearly parallel directions. A tiny change in $\boldsymbol{y}$ can slide $\hat{\boldsymbol{y}}$ along the pancake, dramatically changing the decomposition $(\hat{\beta}_1, \hat{\beta}_2)$ even though $\hat{\boldsymbol{y}} = \hat{\beta}_1\boldsymbol{x}_1 + \hat{\beta}_2\boldsymbol{x}_2$ stays almost the same.

A concrete example. Suppose $x_2 = x_1 + \epsilon$ where ϵ is tiny noise. Then $\hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 \approx (\hat{\beta}_1 + \hat{\beta}_2)x_1$. The model can only estimate the *sum* $\hat{\beta}_1 + \hat{\beta}_2$ reliably, not the individual values. If you run the regression on Monday you might get $(\hat{\beta}_1, \hat{\beta}_2) = (5, 3)$; on Tuesday with slightly different data, you might get $(50, -42)$. Both give approximately the same predictions, but the individual coefficients are meaningless.

The eigenvalue perspective. The eigenvalue decomposition of $\boldsymbol{X}^\top\boldsymbol{X}$ makes this precise:

$$\boldsymbol{X}^\top\boldsymbol{X} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\top, \qquad \boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_p), \quad \lambda_1 \geq \cdots \geq \lambda_p \geq 0.$$

The eigenvalues measure the "width" of the column space in each principal direction. A large eigenvalue λ_k means the column space extends far in direction $\boldsymbol{q}_k$—the projection is stable there. A small eigenvalue λ_k means the column space is paper-thin in that direction—the projection's location is poorly determined.

The variance of the OLS coefficients is $\operatorname{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\boldsymbol{X}^\top\boldsymbol{X})^{-1} = \sigma^2\boldsymbol{Q}\boldsymbol{\Lambda}^{-1}\boldsymbol{Q}^\top$. In the direction of the smallest eigenvalue, the variance is σ^2/λ_p—which explodes when λ_p is small.

The **condition number** $\kappa = \lambda_1/\lambda_p$ measures the aspect ratio of the eigenvalue ellipsoid. When $\kappa \gg 1$, the column space is elongated: the projection onto the column space is stable

(overall R^2 is fine), but the projection's *location within* the thin direction is wildly unstable. Small perturbations in $\boldsymbol{y}$ cause large swings in individual coefficients.

The key danger, summarised. Multicollinearity does *not* make the overall model bad—R^2 and the predicted values $\hat{\boldsymbol{y}}$ may be perfectly fine. What it destroys is the interpretability of individual coefficients. You cannot say "a one-unit increase in x_2 leads to a $\hat{\beta}_2$ change in y, holding x_1 constant" when x_1 and x_2 move together so tightly that "holding x_1 constant" is practically impossible.

Diagnostic: The condition number κ and the eigenvalue spectrum. If $\kappa > 30$, investigate; if $\kappa > 100$, individual coefficients may be unreliable. Another common diagnostic is the **Variance Inflation Factor** (VIF): $\mathrm{VIF}_j = 1/(1 - R_j^2)$, where R_j^2 is the R^2 from regressing predictor x_j on all other predictors. $\mathrm{VIF}_j > 10$ is often flagged as concerning.

7.2 Omitted variable bias: the geometry can't warn you

Warning

Multicollinearity is a problem the geometry can *see*. Omitted variable bias is a problem the geometry *cannot* see.

Suppose the true model is $\boldsymbol{y} = \boldsymbol{X}_1\boldsymbol{\beta}_1 + \boldsymbol{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$ but we estimate the "short" regression omitting $\boldsymbol{X}_2$:

$$\hat{\boldsymbol{\beta}}_1^{\text{short}} = (\boldsymbol{X}_1^\top \boldsymbol{X}_1)^{-1}\boldsymbol{X}_1^\top \boldsymbol{y} = \boldsymbol{\beta}_1 + \underbrace{(\boldsymbol{X}_1^\top \boldsymbol{X}_1)^{-1}\boldsymbol{X}_1^\top \boldsymbol{X}_2}_{\boldsymbol{\Gamma}}\boldsymbol{\beta}_2 + (\boldsymbol{X}_1^\top \boldsymbol{X}_1)^{-1}\boldsymbol{X}_1^\top \boldsymbol{\varepsilon}.$$

Taking expectations:

$$\boxed{E[\hat{\boldsymbol{\beta}}_1^{\text{short}}] = \boldsymbol{\beta}_1 + \boldsymbol{\Gamma}\boldsymbol{\beta}_2.} \tag{5}$$

The bias $\boldsymbol{\Gamma}\boldsymbol{\beta}_2$ is the gap between projections onto two different column spaces. It exists whenever (i) $\boldsymbol{X}_2$ is correlated with $\boldsymbol{X}_1$ ($\boldsymbol{\Gamma} \neq \boldsymbol{0}$) *and* (ii) $\boldsymbol{X}_2$ affects $\boldsymbol{y}$ ($\boldsymbol{\beta}_2 \neq \boldsymbol{0}$).

Both the short and long regressions have perfectly orthogonal residuals, healthy condition numbers, and green diagnostics. The geometry worked flawlessly for both projections. The problem is not how you project—it is *which wall* you project onto. Choosing the right column space requires *causal reasoning*, which lies beyond the geometric framework.

8 Constraining the Shadow: Regularisation

When the column space is nearly degenerate, the OLS projection is "optimal" but unstable. Regularisation deliberately introduces bias to gain stability.

8.1 Ridge regression

Definition 8.1. The **Ridge estimator** with penalty $\lambda \geq 0$ is

$$\hat{\boldsymbol{\beta}}_{\text{ridge}} = \arg\min_{\boldsymbol{\beta}} \left\{ \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|^2 \right\} = (\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I})^{-1}\boldsymbol{X}^\top \boldsymbol{y}.$$

Ridge adds $\lambda \boldsymbol{I}$ to $\boldsymbol{X}^\top \boldsymbol{X}$, which shifts every eigenvalue $\lambda_k \mapsto \lambda_k + \lambda$. The eigendecomposition gives

$$\hat{\boldsymbol{\beta}}_{\text{ridge}} = \boldsymbol{Q}\ \mathrm{diag}\left(\frac{\lambda_k}{\lambda_k + \lambda}\right)\boldsymbol{Q}^\top \hat{\boldsymbol{\beta}}_{\text{OLS}}. \tag{6}$$

Each eigenvector component is shrunk by the factor $\lambda_k/(\lambda_k+\lambda)$. Directions with *large* λ_k (stable directions) are barely affected. Directions with *small* λ_k (unstable directions) are aggressively shrunk toward zero.

Key Geometric Insight

Ridge performs *selective geometric surgery*: it identifies the unstable directions of the column space via the eigenvalue spectrum and shrinks those directions most aggressively, while leaving the stable directions nearly intact.

Geometric interpretation. The OLS solution must lie in the unconstrained column space. The Ridge solution must lie inside a sphere $\{\boldsymbol{\beta} : \|\boldsymbol{\beta}\| \leq t\}$ for some t determined by λ. If the OLS solution is inside the sphere, Ridge and OLS agree. Otherwise, the Ridge solution is the point on the sphere's surface that minimises the residual sum of squares—where the RSS contour ellipse first touches the sphere.

8.2 LASSO regression

Definition 8.2. The **LASSO estimator** with penalty $\lambda \geq 0$ is

$$\hat{\boldsymbol{\beta}}_{\text{lasso}} = \arg\min_{\boldsymbol{\beta}} \left\{ \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|_1 \right\},$$

where $\|\boldsymbol{\beta}\|_1 = \sum_j |\beta_j|$.

The L^1 constraint region is a diamond (cross-polytope) instead of a sphere. Its surface has *corners*—points where one or more coordinates are exactly zero. The RSS contour ellipses are more likely to first touch the diamond at a corner, producing **exact zeros** in the coefficient vector.

Sparsity as a corner solution. LASSO performs automatic variable selection: as λ increases, coefficients are driven to zero one by one. Each zero corresponds to the solution hitting a corner of the L^1 diamond. This is not an algebraic trick—it is a consequence of the geometry of touching a polytope with an ellipsoid.

8.3 Comparison

	Ridge (L^2)	**LASSO (L^1)**
Constraint shape	Sphere (smooth)	Diamond (corners)
Coefficient path	Smooth shrinkage toward 0	Coefficients hit 0 exactly
Variable selection	No (all coefficients remain)	Yes (sparsity)
Best when	Many small effects (dense)	Few large effects (sparse)

Neither Ridge nor LASSO is an orthogonal projection. Both trade a small amount of bias for a (potentially large) reduction in variance.

9 Testing with Geometry: The F-Test

9.1 Comparing two projections

Consider two nested models:

- **Restricted:** $\boldsymbol{y} = \boldsymbol{X}_R\boldsymbol{\beta}_R + \boldsymbol{e}_R$, with column space $\mathcal{C}(\boldsymbol{X}_R)$.
- **Unrestricted:** $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{e}$, with column space $\mathcal{C}(\boldsymbol{X}) \supseteq \mathcal{C}(\boldsymbol{X}_R)$.

Both models project $\boldsymbol{y}$ onto their respective column spaces. The restricted projection $\hat{\boldsymbol{y}}_R$ and the unrestricted projection $\hat{\boldsymbol{y}}$ satisfy $\|\boldsymbol{e}_R\| \geq \|\boldsymbol{e}\|$ (by Proposition **??**). The gap $\hat{\boldsymbol{y}} - \hat{\boldsymbol{y}}_R$ lies in $\mathcal{C}(\boldsymbol{X})$ and is perpendicular to $\mathcal{C}(\boldsymbol{X}_R)$.

9.2 The F-statistic

Definition 9.1. The F-statistic for testing H_0: the q extra predictors have zero coefficients is

$$F = \frac{(\mathrm{SSE}_R - \mathrm{SSE})/q}{\mathrm{SSE}/(n-p)} = \frac{\|\hat{\boldsymbol{y}} - \hat{\boldsymbol{y}}_R\|^2/q}{\|\boldsymbol{e}\|^2/(n-p)}. \tag{7}$$

Under H_0 and standard Gaussian assumptions, $F \sim F_{q,\,n-p}$.

Key Geometric Insight

The F-test measures whether the *distance between two projections* is surprisingly large relative to the residual noise. The numerator is the squared length of the gap between projections, normalised by the number of extra dimensions (q). The denominator is the residual noise level. If $F \approx 1$, the extra predictors captured only noise. If $F \gg 1$, they captured real signal.

Remark 9.2. When $q = 1$ (testing a single coefficient), $F = t^2$ where t is the usual t-statistic. The t-test is the F-test in one dimension.

9.3 Confidence ellipsoids

The sampling distribution of $\hat{\boldsymbol{\beta}}$ is $\hat{\boldsymbol{\beta}} \sim \mathcal{N}(\boldsymbol{\beta},\ \sigma^2(\boldsymbol{X}^\top\boldsymbol{X})^{-1})$. A $(1-\alpha)$-confidence ellipsoid for $\boldsymbol{\beta}$ is

$$\{\boldsymbol{b} \in \mathbb{R}^p : (\hat{\boldsymbol{\beta}} - \boldsymbol{b})^\top(\boldsymbol{X}^\top\boldsymbol{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{b}) \le p\, s^2\, F_{p,n-p,\alpha}\}.$$

The ellipsoid's axes align with the eigenvectors of $\boldsymbol{X}^\top\boldsymbol{X}$ and its semi-axis lengths are proportional to $1/\sqrt{\lambda_k}$. Directions where the column space is "thin" (small λ_k) produce long axes—more uncertainty. This is the same eigenvalue structure that drives multicollinearity and Ridge shrinkage: everything connects.

10 The Limits of the Geometric Framework

The Projection Theorem is powerful but not omniscient. Three fundamental limitations bound the geometric framework.

10.1 Causality

The geometry tells you *how well* you are projecting. It cannot tell you *whether you are projecting onto the right wall.* Omitted variable bias (§**??**) is invisible to every geometric diagnostic. The Scoreboard reads green for a perfectly projected regression on the wrong column space.

To choose the right wall, you need *causal reasoning*: directed acyclic graphs, potential outcomes, natural experiments, instrumental variables. These are structural tools, not geometric ones.

10.2 The inner product

Everything in these notes assumed the Euclidean inner product—all observations weighted equally. If observations have **heteroscedastic** errors ($\mathrm{Var}(\varepsilon_i) = \sigma_i^2$) or are **autocorrelated**, the inner product is wrong. The "perpendicular" is not truly perpendicular in the correct metric, and the "closest point" is not truly closest.

Generalised Least Squares (GLS) fixes this by replacing the Euclidean inner product $\langle \boldsymbol{u}, \boldsymbol{v}\rangle = \boldsymbol{u}^\top\boldsymbol{v}$ with $\langle \boldsymbol{u}, \boldsymbol{v}\rangle_\Omega = \boldsymbol{u}^\top\boldsymbol{\Omega}^{-1}\boldsymbol{v}$, where $\boldsymbol{\Omega} = \mathrm{Var}(\boldsymbol{\varepsilon})$. The projection theorem still applies, but in a different geometry:

$$\hat{\beta}_{\mathrm{GLS}} = (\boldsymbol{X}^\top\boldsymbol{\Omega}^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{\Omega}^{-1}\boldsymbol{y}.$$

10.3 Linearity

The column space is *flat*. If the true relationship is curved, the best flat projection may be a poor representation. Polynomial features enlarge the column space to include some curvature, but the bias-variance trade-off (Notebook 3's overfitting trap) limits how far this can go. Kernel methods, trees, and neural networks define curved "column spaces"—function spaces beyond the reach of linear projection geometry.

11 Summary: The Geometric Rosetta Stone

Every concept in OLS regression has a geometric face and an algebraic face. Table **??** provides the correspondence.

Table 1: The Geometric Rosetta Stone.

Statistical concept	**Geometric interpretation**	**Formula**
Fitted values $\hat{\boldsymbol{y}}$	Shadow on the wall	$\boldsymbol{H}\boldsymbol{y}$
Residuals $\boldsymbol{e}$	Perpendicular gap	$(\boldsymbol{I}-\boldsymbol{H})\boldsymbol{y}$
Coefficients $\hat{\boldsymbol{\beta}}$	Coordinates of the shadow	$(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{y}$
Normal equations	Orthogonality condition	$\boldsymbol{X}^\top\boldsymbol{e}=\boldsymbol{0}$
SST = SSR + SSE	Pythagorean theorem	$\lVert\boldsymbol{y}\rVert^2=\lVert\hat{\boldsymbol{y}}\rVert^2+\lVert\boldsymbol{e}\rVert^2$
R^2	Squared cosine of angle	$\cos^2\theta$
Gauss–Markov (BLUE)	Closest point in subspace	Projection Theorem
Hat matrix $\boldsymbol{H}$	Projection operator	$\boldsymbol{H}^2=\boldsymbol{H},\ \boldsymbol{H}^\top=\boldsymbol{H}$
Leverage h_{ii}	Arm length in column space	$\mathrm{diag}(\boldsymbol{H})$
Cook's distance D_i	Arm length $\times$ surprise	$\propto h_{ii}\cdot e_i^2$
FWL theorem	Sequential orthogonal projections	$\hat{\boldsymbol{\beta}}_2=(\boldsymbol{X}_2^\top\boldsymbol{M}_1\boldsymbol{X}_2)^{-1}\boldsymbol{X}_2^\top\boldsymbol{M}_1\boldsymbol{y}$
Multicollinearity	Thin column space	$\kappa(\boldsymbol{X}^\top\boldsymbol{X})\gg 1$
OVB	Wrong column space	Bias $=\boldsymbol{\Gamma}\boldsymbol{\beta}_2$
Ridge	Constrained projection (sphere)	Shrink unstable eigenvalue directions
LASSO	Constrained projection (diamond)	Corner solutions $\Rightarrow$ sparsity
F-test	Gap between two projections	$F=\frac{\lVert\hat{\boldsymbol{y}}-\hat{\boldsymbol{y}}_R\rVert^2/q}{\lVert\boldsymbol{e}\rVert^2/(n-p)}$
Confidence ellipsoid	Region of plausible shadows	Axes $\propto 1/\sqrt{\lambda_k}$

Regression is a projection. Coefficients are shadows. The residual is perpendicular. R^2 is an angle. Variance decomposition is Pythagoras. Gauss–Markov is "closest point." Controlling for is residualising. Leverage is arm length. Influence is leverage times surprise. Multicollinearity is a thin column space. The F-test compares two projections. Ridge shrinks the unstable directions. LASSO exploits diamond corners. And the geometry cannot tell you which wall to aim at.

The geometry was always there. These notes gave you the flashlight.

Further reading.

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